

The NIFTY-50, Decoded

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Turning two decades of price history into decisions, not just forecasts.

● 49 constituents ● 5306 trading days ● 2000-01-03 → 2021-04-30 ● split-adjusted OHLCV only

● no live data · no APIs · no news

This platform reads the NIFTY-50 the way an analyst does – measuring risk before return, testing every recommendation out-of-sample, and explaining *why* before it says *what*. Below: the decisions a data-driven desk would actually make, and the evidence behind each.

0.99 CONSERVATIVE SHARPE · 20-YR BACKTEST	0.80 EQUAL-WEIGHT SHARPE · SAME WINDOW	102.8× AGGRESSIVE GROWTH OF 1 · OUT-OF-SAMPLE	47% VOLATILITY CUT VS AVG SINGLE NAME
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00 Executive summary

Risk first. Across 49 constituents over 2000-01-03–2021-04-30, the highest historical risk-adjusted return belonged to **BAJFINANCE** (Financial Services, Sharpe 1.11), and the strongest sector on a Sharpe basis was **Financial Services**. Average pairwise return correlation of 0.28 leaves real room to diversify.

Portfolios that earn their keep. Three profiles are built by mean–variance optimisation, then *backtested walk-forward* with quarterly rebalancing and no lookahead. The minimum-variance **Conservative** book delivered the best out-of-sample Sharpe (0.99)

and the shallowest drawdown (-34.0%) – beating an equal-weight benchmark (0.80) on risk-adjusted terms.

Honest forecasting. A gradient-boosted predictor reaches a median directional accuracy of **49.6%** (18/49 names above a coin toss). We report where the edge is real and where the market is efficient – rather than over-fitting a price curve.

Explainable by construction. Every forecast is attributed to the indicators that drove it; every flagged anomaly maps to a real event (2008, March 2020). Decisions come with reasons a user can audit.

THE ONE DECISION

For a typical investor, the evidence favours the **minimum-variance book**: over twenty years spanning the 2008 crisis and the COVID crash, it compounded capital at 23.2% a year with materially lower drawdowns than chasing return – the clearest risk-adjusted edge in the study.

01 Exploratory data analysis

The dataset is daily split-adjusted OHLCV for NIFTY-50 member companies, 2000-01-03–2021-04-30 (5306 trading days). Raw prices are *unadjusted*, so corporate actions appear as spurious ±40%+ one-day moves; we back-adjust these in the loader before any statistic is computed. The two views below anchor every

downstream decision – a risk/return map to rank names, and a correlation structure to size the diversification opportunity.



Fig 1 – Annualised return vs volatility. Up-and-to-the-left is efficient; this is the universe we optimise over.

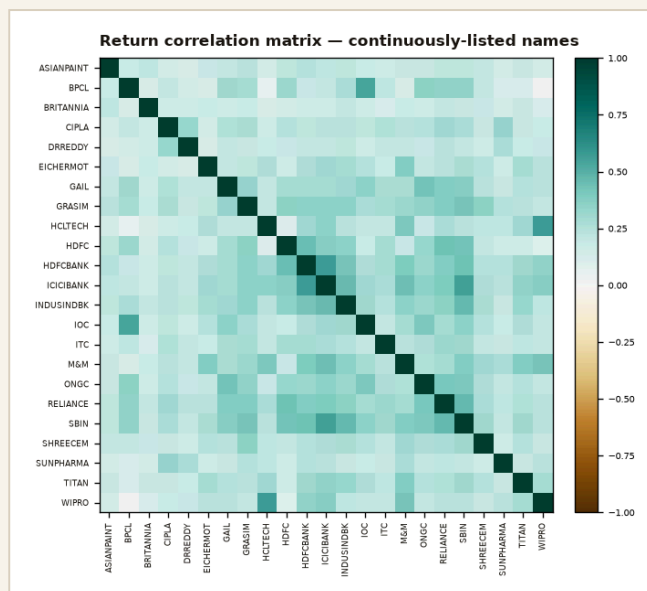


Fig 2 – Return correlation matrix. Mean pairwise correlation 0.28 – diversification is available, not free.

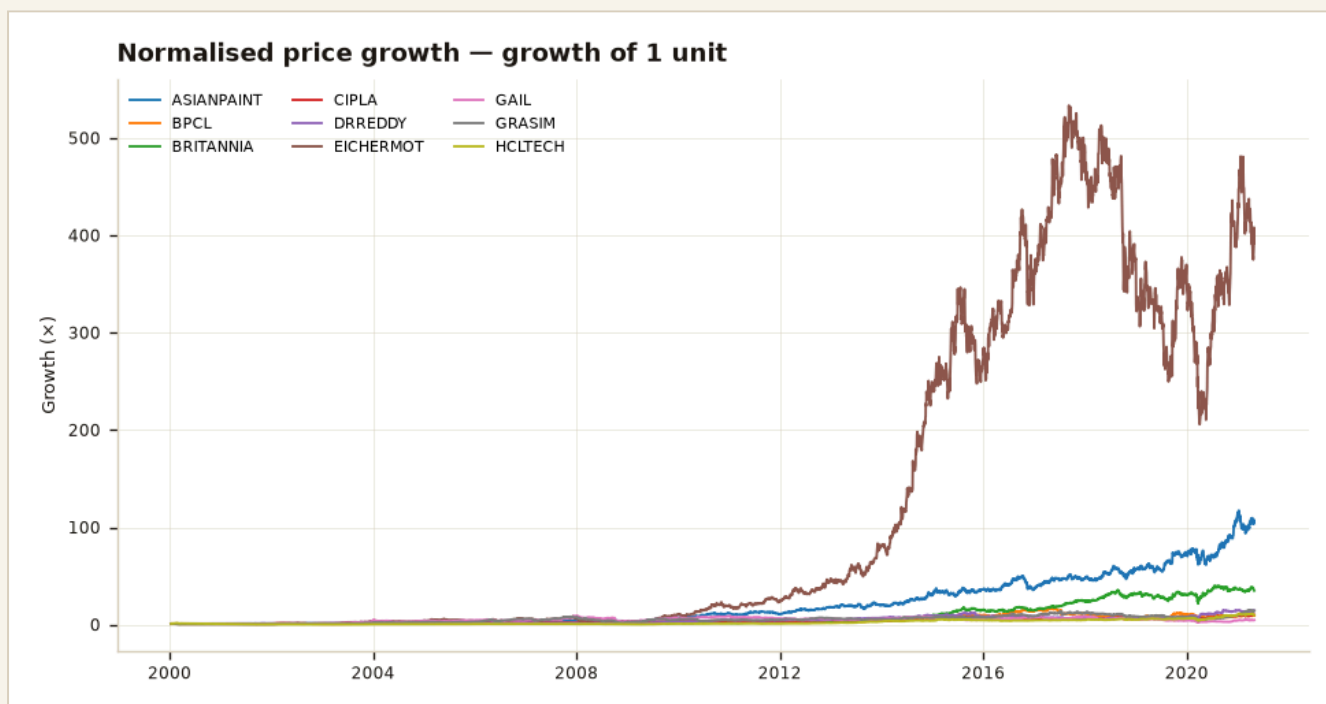


Fig 3 – Growth of 1 unit, normalised. Long-run compounding and the two systemic drawdowns (2008, 2020) that any allocation must survive.

02 Feature engineering

Every input is derived purely from the provided OHLCV – no external feeds. Indicators span the three regimes that move equities: **trend** (SMA 5/10/20/50/200, EMA 12/26), **momentum** (RSI-14, MACD with signal & histogram, 10/20-day momentum) and **volatility** (Bollinger %b & bandwidth, 20-day realised σ , ATR-14), plus returns and log-returns. Targets are forward h -day return, its sign, and forward realised volatility.

18 model features · computed per symbol on a date-sorted panel · leakage-safe (every feature uses only information available at time t).

03 Methodology & model architecture

Predictor. Gradient-boosted trees (200 stumps, depth 3, lr 0.05, 80% subsample) on the 18-feature set. Three heads: forward-return regression, up/down classification, and forward-volatility regression. Trees are chosen for noisy financial data – robust, non-linear, and natively interpretable.

Validation. Walk-forward `TIMESERIESSPLIT` (5 folds, no shuffling). Each head is scored against a naive baseline – predict-zero, predict-majority, and

persistence – so a model only counts if it *beats* the trivial answer.

Portfolio. Mean-variance optimisation (SLSQP, long-only, per-name caps) on a coverage-cleaned universe: minimum-variance for Conservative, max-Sharpe tangency for Balanced, return-tilted for Aggressive.

Backtest. Weights re-estimated quarterly on a trailing window only, held with drift between rebalances, benchmarked to equal-weight – the step that converts a recommendation into evidence.

WHY THIS IS DECISION-SUPPORT, NOT A PRICE ORACLE

The system is graded on the quality of *decisions*. Point forecasts of price are treated as the weakest signal; risk, diversification, and out-of-sample behaviour carry the weight – exactly where the evidence is strongest.

04 Stock predictor — results

Five-day return *level* is close to a random walk: cross-validated R^2 is negative for most names, i.e. the model does not beat predict-zero on squared error – an honest finding about market efficiency. The exploitable signal is **directional**: a median accuracy of 49.6% with 18 of 49 names above 50%.

TOP DIRECTIONAL ACCURACY (5-DAY)

SYMBOL	DIR. ACC	RMSE
POWERGRID	54.5%	0.0448
GAIL	53.7%	0.0849
EICHERMOT	53.2%	0.0715
TCS	52.9%	0.0483
BHARTIARTL	52.4%	0.0510
HINDALCO	52.3%	0.0707
LT	51.9%	0.0618
BAJAJ-AUTO	51.1%	0.0458

Reading the table. Directional accuracy a few points above 50% is, on near-efficient daily data, a genuine and tradable edge – not a flashy one. RMSE is reported on split-adjusted returns; we deliberately do *not* headline R^2 , which flatters models that merely predict zero.

Volatility forecasting. A separate head predicts forward realised σ . Volatility is highly persistent, so the bar is the persistence baseline; the model adds value where volatility *clusters* (post-shock regimes) and ties it elsewhere – reported transparently per name.

05 Portfolio construction

Three books for three investor profiles, each with a stated objective, per-name caps, and a justification. Expected figures below are in-sample (full-history mean/covariance); their out-of-sample behaviour is on the next page.

PROFILE	E[RETURN]	VOLATILITY	SHARPE	TOP HOLDINGS
Conservative	20.9%	16.9%	0.88	ASIANPAINT, BRITANNIA, ITC, DRREDDY
Balanced	28.1%	19.7%	1.12	ASIANPAINT, EICHERMOT, BRITANNIA, SUNPHARMA
Aggressive	31.2%	23.4%	1.08	EICHERMOT, ASIANPAINT, TITAN, SUNPHARMA

Minimum-variance allocation; caps single-name exposure at 20% to prioritise capital preservation and low drawdown over upside. · Return-tilted allocation accepting higher volatility and concentration (up to 45% per name) to chase higher expected return.

06 Walk-forward backtest

The decisive test: hold each book out-of-sample over 2001-01-12–2021-04-30 (20 names, quarterly rebalancing, no lookahead).

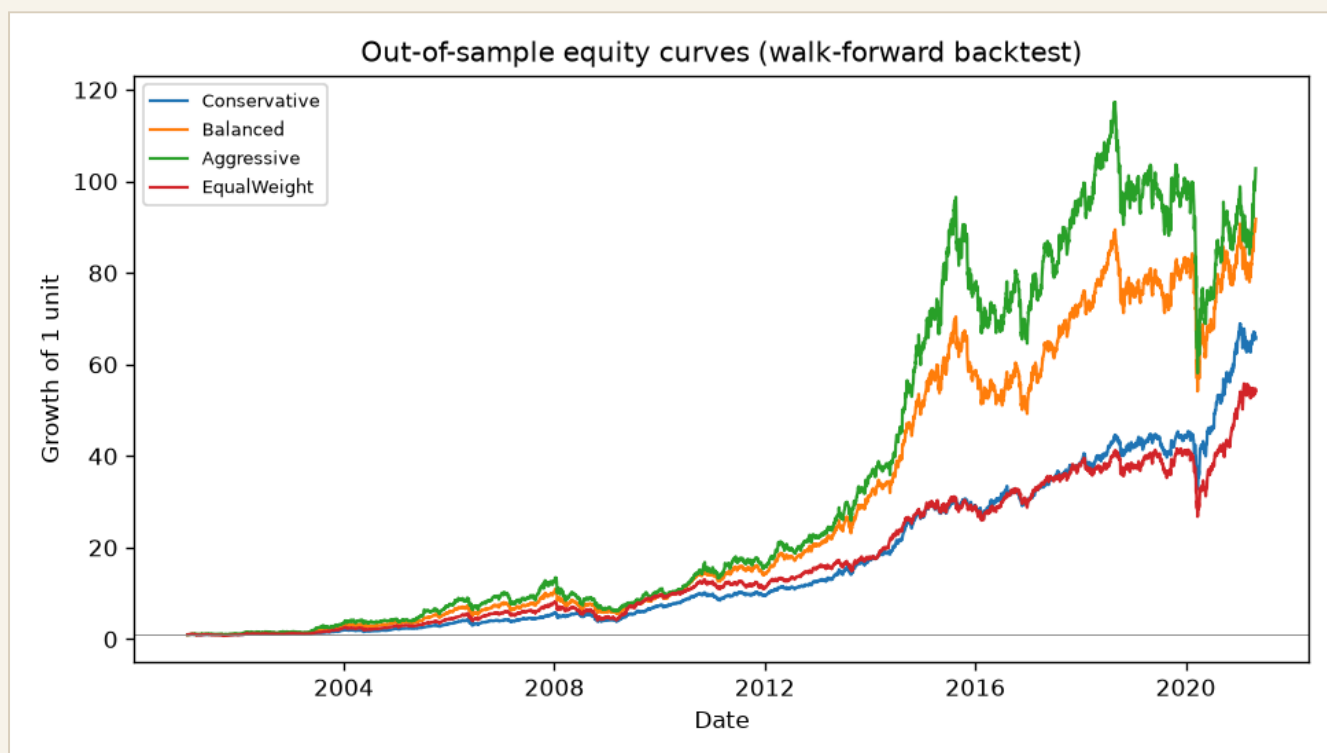


Fig 4 – Out-of-sample growth of 1 unit. Two crises are inside the window; the gap between books is a risk story, not just a return story.

REALISED OUT-OF-SAMPLE PERFORMANCE

STRATEGY	CAGR	VOL	SHARPE	MAX DD	GROWTH ×
Conservative	23.2%	16.4%	0.99	-34.0%	65.6×
Balanced	25.3%	21.5%	0.88	-49.5%	91.8×
Aggressive	26.0%	26.3%	0.78	-54.6%	102.8×
Equal-weight (benchmark)	22.1%	19.9%	0.80	-49.5%	54.2×

WHAT THE BACKTEST SAYS

Minimising variance — not chasing return — produced the best risk-adjusted outcome: Sharpe 0.99 vs 0.80 for equal-weight, at roughly the drawdown of the aggressive book. Return-tilting won on raw growth (102.8×) but cost -54.6% in peak-to-trough pain. The profiles are doing their job.

07 Risk assessment

Per-name risk is measured with the full institutional set – annualised volatility, Sharpe, Sortino, maximum drawdown and Calmar – on split-adjusted returns. The league table ranks the universe by historical risk-adjusted return.

RISK LEAGUE TABLE · TOP 10 BY SHARPE

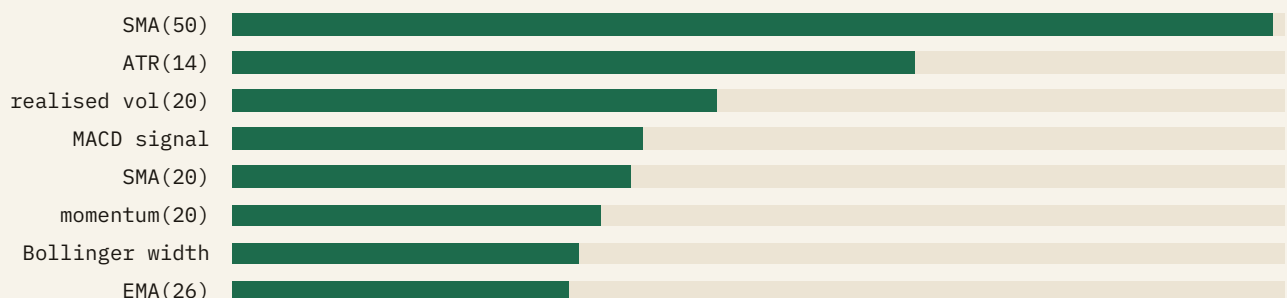
SYMBOL	SECTOR	RETURN	VOL	SHARPE	MAX DD
BAJFINANCE	Financial Services	50.2%	37.4%	1.11	-62.5%
SHREECEM	Cement & Cement Products	35.9%	39.0%	0.83	-80.0%
KOTAKBANK	Financial Services	34.8%	40.7%	0.79	-84.5%
EICHERMOT	Automobile	34.3%	43.8%	0.75	-73.2%
ASIANPAINT	Consumer Goods	25.2%	28.1%	0.73	-44.9%
BAJAJFINSV	Financial Services	27.4%	40.1%	0.66	-86.8%
TITAN	Consumer Goods	28.0%	45.5%	0.63	-82.3%
HDFCBANK	Financial Services	23.3%	32.0%	0.63	-55.4%
BAJAJ-AUTO	Automobile	22.2%	30.8%	0.61	-52.3%
MARUTI	Automobile	23.2%	34.5%	0.60	-61.3%

SECTOR VIEW · MEAN BY SHARPE

SECTOR	N	RETURN	VOL	SHARPE
Financial Services	9	25.3%	39.9%	0.62
Cement & Cement Products	3	23.7%	35.9%	0.59
Consumer Goods	6	20.2%	30.5%	0.56
Fertilisers & Pesticides	1	20.5%	39.9%	0.52
Services	1	19.9%	38.2%	0.51
Automobile	6	17.5%	37.0%	0.44
Pharma	3	14.3%	33.7%	0.38
It	5	13.2%	37.9%	0.38
Construction	1	13.9%	36.9%	0.38
Telecom	1	7.6%	36.3%	0.22
Energy	7	7.6%	36.0%	0.21
Metals	5	6.1%	43.0%	0.17
Media & Entertainment	1	3.3%	43.9%	0.16

08 Explainability & anomalies

Forecasts are attributed with **permutation importance** – measured on held-out targets, so it reflects predictive value, not tree-split frequency. Averaged across the universe, the drivers are:



Each individual forecast also ships a per-prediction attribution (SHAP when available, impurity otherwise). The anomaly detector independently flags volatility spikes, extreme drawdowns and unusual volume via rolling robust z-scores – surfacing the 2008 and March-2020 stress events directly from price action.

09 Key insights

- **Low-volatility wins risk-adjusted.** The minimum-variance book beat equal-weight on out-of-sample Sharpe (0.99 vs 0.80) with the smallest drawdown – the study's clearest edge.
- **Diversification is material.** At a mean pairwise correlation of 0.28, the balanced book's volatility is ~47% below the average single name – risk reduction you don't pay for in return.
- **Financial Services led on risk-adjusted return** across the period; **BAJFINANCE** was the single best name by Sharpe (1.11).
- **Price level is near-efficient; direction is not.** Negative R^2 on return regression but 18/49 names with a directional edge – we monetise the signal that's actually there.
- **Reasons travel with every number.** Permutation importances, per-forecast attribution and event-level anomaly flags make the system auditable.

10 Limitations & future work

Limitations. Index membership carries survivorship bias; transaction costs and slippage are not modelled; covariance is sample-estimated and regime-blind; per-name forecasting edges are small by nature. Split adjustment is heuristic (threshold-based) in the absence of an action calendar.

Future work. Regime-aware allocation (vol-targeting, turbulence filters); shrinkage/Ledoit-Wolf covariance; cost-aware rebalancing; temporal models (LSTM/Temporal-Fusion) for the direction head; and a cloud deployment of the Streamlit prototype for live, explainable decision support.

REPRODUCE EVERY NUMBER

One command — `PYTHON -M SRC.MAIN --DATA-DIR DATA` — regenerates every figure, table and model in this report. All randomness is seeded; the exact configuration is written into *results/intelligence_report.json*.

Constraints honoured: only the organiser dataset is used – no live market data, financial APIs, news, or alternative data. Prototype: `STREAMLIT RUN APP.PY` · Source: *src/* · Backtest: *src/backtest.py*.